Seongjae Kang

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Seoul, South Korea

RESEARCH INTERESTS

AI systems must account for human requirements that may be implicit, differ across stakeholders, or change through use. I am interested in understanding and representing these requirements so they can guide model learning and system behavior while remaining open to human revision.

- **Understanding and representing human requirements**: Inferring candidate requirements from partial instructions, demonstrations, corrections, and interaction histories, retaining the evidence and scope supporting an interpretation, and representing requirements in forms people can inspect and correct.
- **Adapting learning and behavior**: Translating requirements into objectives and constraints, carrying revisions through model learning and system behavior, and evaluating whether intended changes take effect while continuing to satisfy applicable requirements and preserving human authority over consequential tradeoffs.

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology) **Feb 2022 – Feb 2024**
M.S. in Graduate School of AI *Seoul, South Korea*
Advisor: Prof. Sung Ju Hwang
GPA: 3.96/4.3

KAIST (Korea Advanced Institute of Science and Technology) **Feb 2016 – Feb 2022**
B.S. in Computer Science (major) and Electrical Engineering (minor) *Daejeon, South Korea*
GPA: 3.88/4.3
Korea National Science & Technology Scholarship, full tuition, 2018 – 2021

EXPERIENCE

Match Group (MG AI) **May 2026 – Present**
Machine Learning Engineer (Military Alternative Service) *Seoul, South Korea*

- **Recommendation Systems:** Developing machine learning models to improve recommendation and matching quality across the platform
- **Agentic Systems:** Contributing to agentic systems for real-world applications, with a focus on policy adherence and safe workflow execution

VUNO **May 2024 – May 2026**
Research Scientist/Engineer (Military Alternative Service) *Seoul, South Korea*

- **Medical Image Analysis:** Developed deep learning models for detecting findings and diseases in chest X-ray images, investigating methods to leverage chest X-ray foundation models' knowledge for compact deployment-ready models
  - Developed methods for data-efficient learning approaches including few-shot learning and semi-supervised learning, tailored to the characteristics of specific findings and diseases
  - Conducted research on knowledge distillation framework to transfer large vision-language medical foundation models into compact models suitable for clinical deployment

- **Human-in-the-Loop Data Acquisition:** Designed and implemented active learning pipeline to address the high labeling costs of medical imaging data
  - Developed full-stack proof-of-concept data engine integrating Label Studio and MongoDB for efficient annotation workflows
  - Conducted research on active learning frameworks and selection method under the regime of foundation model knowledge distillation
- **Automated Radiology Report Generation:** Developed AI systems to assist radiologist workflows through automated report generation
  - Investigated how recent Multimodal Large Language Models (MLLMs) incorporate clinical contexts (e.g., indication, clinical history) when generating radiology reports
  - Developed proof-of-concept demonstration for keyword-based radiology report generation to assist radiologist workflows using MLLMs and Gradio UI
- **Workflow Automation:** Voluntarily developed automated cash reimbursement system that automatically fills submission forms, reducing processing time by approximately 1 hour per month for each employee
  - Implemented JavaScript-based web automation for form interaction and navigation
  - Designed user interface and distributed solution via Chrome Extension Store

Deepauto AI **Feb 2024 – Mar 2024**
Research Scientist/Engineer *Seoul, South Korea*

- **Universal Vision Task Solver:** Investigated how multimodal large language models (MLLMs) could solve multiple computer vision tasks within a shared framework
  - Reformulated image classification, object detection, and image segmentation as an instruction-tuning dataset
  - Formulated VQ-VAE image tokens and integrated them into LLaVA to support image outputs

VoyagerX **Sep 2020 – Sep 2021**
Research Engineer (Internship) *Seoul, South Korea*

- **Document Image Enhancement:** Developed lightweight deep learning models for enhancing scanned documents and removing fingers from document images
  - Collected document images across different lighting conditions and materials, along with finger images spanning different postures and demographics
  - Generated synthetic training data for document enhancement and finger removal
  - Deployed the models with TensorFlow Lite and integrated them into an Android application using Java

SGVR Lab, KAIST **Mar 2019 – Aug 2019**
Undergraduate Research Assistant *Daejeon, South Korea*
Advisor: Prof. Sung-eui Yoon

- **Vision-Based Virtual Piano:** Developed a computer vision system that converted performers' hand movements into piano output
  - Collected videos of multiple participants pressing piano keys
  - Designed a spatiotemporal model to detect key locations and estimate key-press intensity
  - Integrated MIDI synthesis to generate audio from the model's predictions

Computer Vision Lab, KAIST **Jun 2017 – Jun 2018**
Undergraduate Research Assistant *Daejeon, South Korea*
Advisor: Prof. Hyunseung Yang

- **Computer Vision Research:** Studied computer vision and deep learning methods

AWARDS & HONORS

Korea National Science & Technology Scholarship **2018 – 2021**
Ministry of Science and ICT, Republic of Korea
National merit scholarship covering full tuition.
The 2018 nationwide selection target was approximately 3,350 students.

1st Prize, Semantic Segmentation AI Model for Body Parts Challenge **2021**
Alchera, AI Heroes *South Korea*

Linux Master 1st Grade Certificate
Korea Information Technology Human Resources Development Institute *South Korea*

ACADEMIC SERVICE

Conference Reviewer **2026**
ICML (Gold Reviewer), NeurIPS, AAAI

PUBLICATIONS

*(P: preprint, C: conference, J: journal, W: workshop, *: equal contribution)*

- **[C1] Simple yet Effective Semi-supervised Knowledge Distillation from Vision-Language Models via Dual-Head Optimization**
  Seongjae Kang*, Dong Bok Lee*, Hyungjoon Jang, Sung Ju Hwang
  Advances in Neural Information Processing Systems (NeurIPS), 2026

- **[C2] Deconstructing Multi-Task Active Learning: The Paradox of Gradient Conflict and Orthogonal Decomposition**
  Juho Jung*, **Seongjae Kang***, Jinyoung Han
  Advances in Neural Information Processing Systems (NeurIPS), 2026

- **[C3] Frequency-Balanced Retinal Representation Learning with Mutual Information Regularization**
  Seunghoon Lee, **Seongjae Kang**, Inhyuk Park, Gitaek Kwon, Jihyeon Baek, Doohyun Park
  International Conference on Learning Representations (ICLR), 2026

- **[C4] High-Suspicion Pulmonary Nodule Detection on Chest Radiographs: Single-Center Retrospective Performance and Additional Detection Analysis of an AI System**
  Y. Hwang, S. Ham, Y. Jun, J. Kang, S. Cho, **S. Kang**, S. H. Lee
  European Congress of Radiology (ECR), Scientific Exhibit, 2026

- **[C5] CostNav: A Navigation Benchmark for Real-World Economic-Cost Evaluation of Physical AI Agents**
  Haebin Seong, Sungmin Kim, Yongjun Cho, Myunchul Joe, Geunwoo Kim, Yubeen Park, Sunhoo Kim, Samwoo Seong, Yoonshik Kim, Suhwan Choi, Jaeyoon Jung, Jiyong Youn, Jinmyung Kwak, Sunghee Ahn, Jaemin Lee, Younggil Do, Seungyeop Yi, Woojin Cheong, Minhyeok Oh, Minchan Kim, **Seongjae Kang**, Youngjae Yu, Yunsung Lee
  Conference on Robot Learning (CoRL), 2026; also presented at the ScaleBot Workshop at CVPR, 2026

- **[W1] PolicyGuide: From Guarding One Action to Guiding the Whole Workflow for Policy-Compliant LLM Agents**
  Seongjae Kang, Taehyung Yu, Sung Ju Hwang
  Who Verifies the Agents? Toward Reliable Agent Development (NeurIPS 2026 Workshop), 2026

- **[W2] PolicyGuard: A Dialogue-Grounded Sub-Agent Verifier for Policy Adherence in LLM Agents**
  Seongjae Kang, Taehyung Yu, Sung Ju Hwang
  Who Verifies the Agents? Toward Reliable Agent Development (NeurIPS 2026 Workshop), 2026

- **[W3] Best-of-N TTS Evaluation is Confounded by ASR Family Alignment**
  Taehyung Yu, **Seongjae Kang**
  Machine Learning for Audio Workshop at ICML, 2026

- **[P1] Automated Structured Radiology Report Generation with Rich Clinical Context**
  Seongjae Kang*, Dong Bok Lee*, Juho Jung, Dongseop Kim, Won Hwa Kim, Sunghoon Joo
  Under review at NAACL 2027

- **[P2] PCoreSet: Effective Active Learning through Knowledge Distillation from Vision-Language Models**
  Seongjae Kang*, Dong Bok Lee*, Hyungjoon Jang, Dongseop Kim, Sung Ju Hwang
  Under review at CVPR 2027

- **[P3] Virtual Piano using Computer Vision**
  Seongjae Kang, Jaeyoon Kim, Sung-eui Yoon
  arXiv preprint arXiv:1910.12539, 2019

SELECTED PROJECTS

ScreenTranslate — On-Device Screen Translation for macOS
Tools: Swift, macOS Screen Capture, Vision OCR, Apple Translation

- Built a native macOS menu-bar app that translates text in images, PDFs, applications, and videos through a keyboard shortcut
  - Combined screen capture, on-device OCR, and Apple's Translation framework in one workflow
  - Added area-selection, text-selection, and quick-type modes with offline support for 20 languages
  - Added optional DeepL, Google, and Azure engines for broader language coverage
- Released the app as free and open-source software under the GPL-3.0 license

YouTube Multi-Subtitle for Chinese Learning
Tools: JavaScript, Chrome Extension API

- Built a web application for learning Chinese through YouTube videos with simultaneous display of three subtitle tracks
  - Recognized that Chinese requires both characters and Pinyin (pronunciation) to learn effectively, unlike other languages where a single subtitle suffices
  - Implemented real-time Chinese-to-Pinyin conversion to automatically generate pronunciation guides
  - Enabled loading of English and Chinese subtitles from multiple sources including machine translation
  - Replaced YouTube's default subtitle with custom overlay showing English meaning, Chinese characters, and Pinyin together
- Made Chinese learning significantly easier by providing all necessary linguistic information in one place

Easy - Expense Report Automation Chrome Extension
Tools: JavaScript, Chrome Extension API

- Developed a Chrome extension that automates the monthly expense settlement process for company employees
  - Recognized that monthly expense reporting involves repetitive form filling and data entry tasks that consume significant time
  - Implemented JavaScript-based web automation for automated form interaction, data extraction, and navigation
  - Designed intuitive user interface to streamline the workflow and minimize user intervention
  - Distributed the solution via Chrome Extension Store to make it accessible to all company employees
- Reduced what typically takes 1 hours of manual work each month into just a few minutes with minimal effort, significantly improving productivity and reducing repetitive administrative tasks

SKILLS

Languages: Korean (Native), English (Fluent), Spanish (● ● ○), Chinese (● ○ ○)
Programming Languages: Python, Java, C, JavaScript, and English*
Interests: Continuous Self-improvement, Language Learning, Swimming

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